1.1 QUBIT (Quantum Bit)

A qubit, short form of "quantum bit," is a fundamental unit of information in quantum computing. It is the quantum analog of a classical bit, which is represented as 0 or 1 in classical computing.

A qubit is a two-level quantum system where the two basis qubit states are usually written as |0⟩ and |1⟩. A qubit can be in state |0⟩ or |1⟩ or in a linear combination of both states.

When we discuss qubit basis states, we essentially alluding to the z-basis states as the fundamental computational states.

Properties of qubits

1. Qubits make use of discrete energy state particles such as electrons and photons.
2. Qubits exists in two quantum state |0⟩ and |1⟩ or in a linear combination of both states. This is known as superposition.
3. Unlike classical bits, qubit can work with the overlap of both 0 & 1 states.
4. When the qubit is measured, it collapses to one of the two basis states |0⟩ or |1⟩
5. Quantum entanglement and quantum tunnelling are two exclusive properties of qubit.
6. State of the qubits is represented using Bloch sphere.

Example :

Suppose there are two basis Qubit states $|0\rangle$ ***and*** $|1\rangle$

These states are such that

$\langle 0|0\rangle = \langle 1|1\rangle = 1$ ***(Normalized basis)***

$\langle 0|1\rangle = \langle 1|0\rangle = 0$ ***(orthogonal)***

Any linear combination of these two basis states will also be a basis states.

$$|\psi\rangle = 0.5|0\rangle + 0.2i|1\rangle$$

Or in general

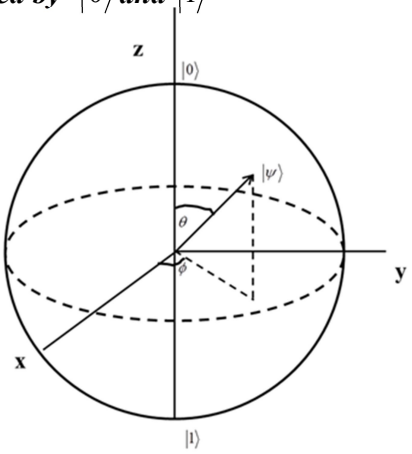
$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Important Note: Where a and b are the complex probability amplitude for respective basis states. **The condition on a and b means that |Ψ⟩ should be normalised.**

$$|a|^2 + |b|^2 = 1$$

Example: suppose basis Qubit states are represented by $|0\rangle$ ***and*** $|1\rangle$

Then it linear combination can also be written as



$$|\psi\rangle = |\theta,\phi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

For θ=0 and ϕ=0

$$|\psi\rangle = |0,0\rangle = \cos(0)|0\rangle + e^{i0}\sin(0)|1\rangle = |0\rangle$$

For $\theta=\frac{\pi}{2}$ and ϕ=0

$$|\psi\rangle = \left|\frac{\pi}{2},0\right\rangle = \cos\left(\frac{\pi}{4}\right)|0\rangle + e^{i0}\sin\left(\frac{\pi}{4}\right)|1\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$$

For $\theta=\pi$ and ϕ=0

$$|\psi\rangle = |\pi,0\rangle = \cos\left(\frac{\pi}{2}\right)|0\rangle + e^{i0}\sin\left(\frac{\pi}{2}\right)|1\rangle = |1\rangle$$

1.2 Qudit:

A Qudit or a d-dimensional quantum system can be represented as d-dimensional ket vector |Ψ⟩

$$|\Psi\rangle = \sum_{i=0}^{d-1} \alpha_i\, |\phi_i\rangle$$

provided

$$\sum_{i=0}^{d-1} |\alpha_i|^2 = 1$$

The condition on the coefficients α_i means that |Ψ⟩ is a vector of length of 1.

1.3 STANDARD BASIS:

Consider a 2 dimensional orthonormal complex vector space. The standard basis also sometimes known as computational basis $\mathcal{S} = \{|0\rangle, |1\rangle\}$, is an orthonormal basis for this vector space, where the basis vectors are

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix} \quad and\ |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

1.4 HADAMARD BASIS:

The Hadamard basis is an orthonormal basis $\mathcal{H} = \{|0\rangle, |1\rangle\}$, is an orthonormal basis for this vector space, where the basis vectors are

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix},\ |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

Let us check this is an orthonormal basis

$$\langle +|-\rangle = \langle -|+\rangle = 0, \& \langle +|+\rangle = \langle -|-\rangle = 1$$

Example: Express $|1\rangle$ in the Hadamard basis. That is, find coefficients α and β such that $|1\rangle = \alpha|+\rangle + \beta|-\rangle$.

Since we know that standard basis or computational basis is defined as

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix} \quad and\ |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

Therefore Hadamard basis can be written as

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}\left(\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix}\right) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}\left(\begin{bmatrix}1\\0\end{bmatrix} - \begin{bmatrix}0\\1\end{bmatrix}\right) = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

$$|1\rangle = \alpha|+\rangle + \beta|-\rangle = \alpha\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} + \beta\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}.$$

$$\Rightarrow \begin{bmatrix}0\\1\end{bmatrix} = \alpha\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} + \beta\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}0\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}\alpha+\beta\\\alpha-\beta\end{bmatrix}$$

$$\Rightarrow\ \alpha + \beta = 0$$

$$\Rightarrow \alpha - \beta = \sqrt{2}$$

$$\Rightarrow \alpha = -\beta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \mathbf{|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)}$$

1.5 MULTIPLE QUBIT

Classically if we have two bits, we write as “00”, “01” and so on.

How can we write two Qubits?

For example: suppose we want to write $|00\rangle$, $|11\rangle$

Since we know that standard basis or computational basis is defined as

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix} \quad and\ |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

Before solving above problem we should learn **Kronecker or Tensor product of two Matrices**

A=$|A\rangle=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ and B=$|B\rangle=\begin{bmatrix}0 & 5\\ 6 & 7\end{bmatrix}$

Therefor tensor product of two matrices can be written as

$$A\otimes B=\begin{bmatrix}1\times\begin{bmatrix}0 & 5\\ 6 & 7\end{bmatrix} & 2\times\begin{bmatrix}0 & 5\\ 6 & 7\end{bmatrix}\\ 3\times\begin{bmatrix}0 & 5\\ 6 & 7\end{bmatrix} & 4\times\begin{bmatrix}0 & 5\\ 6 & 7\end{bmatrix}\end{bmatrix}$$

$$\Rightarrow A\otimes B=\begin{bmatrix}0 & 5 & 0 & 10\\ 6 & 7 & 8 & 14\\ 0 & 15 & 0 & 20\\ 18 & 21 & 24 & 28\end{bmatrix}$$

Let us do the example for how to write the multiple Qubit?

$$A=|0\rangle=\begin{bmatrix}1\\ 0\end{bmatrix}$$

$$B=|1\rangle=\begin{bmatrix}0\\ 1\end{bmatrix}$$

$$A\otimes A=|00\rangle=\begin{bmatrix}1\\ 0\end{bmatrix}.B=\begin{bmatrix}1\times\begin{bmatrix}1\\ 0\end{bmatrix}\\ 0\times\begin{bmatrix}1\\ 0\end{bmatrix}\end{bmatrix}=\begin{bmatrix}1\\ 0\\ 0\\ 0\end{bmatrix}$$

$$B\otimes B=|11\rangle=\begin{bmatrix}0\\ 1\end{bmatrix}.B=\begin{bmatrix}0\times\begin{bmatrix}0\\ 1\end{bmatrix}\\ 1\times\begin{bmatrix}0\\ 1\end{bmatrix}\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\\ 1\end{bmatrix}$$

Example: Consider two qubits A and B, in the two qubit state known as the EPR pair one can label the joint state as AB?

$$|EPR\rangle_{AB}=\frac{1}{\sqrt{2}}(|00\rangle_{AB}+|11\rangle_{AB})=\frac{1}{\sqrt{2}}\left(\begin{bmatrix}1\\ 0\\ 0\\ 0\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 0\\ 1\end{bmatrix}\right)=\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ 0\\ 0\\ 1\end{bmatrix}$$

Where EPR stands for Einstein, Podolsky and Rosen, (we will study later).

Which is equal to the superposition between the vectors $|00\rangle_{AB}\ and\ |11\rangle_{AB}$.

The length of this vector is calculated using inner product.

$$\langle EPR|EPR\rangle_{AB} = \frac{1}{\sqrt{2}}(\langle 00|_{AB} + \langle 11|_{AB}) \times \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}([1 \quad 0 \quad 0 \quad 0] + [0 \quad 0 \quad 0 \quad 1]) \times \left(\begin{bmatrix}1\\0\\0\\0\end{bmatrix} + \begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right)$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}\left([1 \quad 0 \quad 0 \quad 0] \times \begin{bmatrix}1\\0\\0\\0\end{bmatrix} + [1 \quad 0 \quad 0 \quad 0] \times \begin{bmatrix}0\\0\\0\\1\end{bmatrix} + [0 \quad 0 \quad 0 \quad 1] \times \begin{bmatrix}1\\0\\0\\0\end{bmatrix} + [0 \quad 0 \quad 0 \quad 1] \times \begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right)$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}(\langle 00|00\rangle_{AB} + \langle 00|11\rangle_{AB} + \langle 11|00\rangle_{AB} + \langle 11|11\rangle_{AB})$$

$$\Rightarrow \langle EPR|EPR\rangle_{AB} = \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

1.6 How to combine Qubits?

Let us consider we have two Qubits A and B. We describe the state A as $|\Psi\rangle_A$ and state B as $|\Phi\rangle_B$.

$$|\Psi\rangle_A = \alpha_A |0\rangle_A + \beta_A |1\rangle_A$$

$$|\Phi\rangle_B = \alpha_B |0\rangle_B + \beta_B |1\rangle_B$$

Tensor product of two vectors

$$|\Psi\rangle_{AB} = |\Psi\rangle_A \otimes |\Phi\rangle_B = \begin{bmatrix}\alpha_A\alpha_B\\ \alpha_A\beta_B\\ \beta_A\alpha_B\\ \beta_A\beta_B\end{bmatrix}$$

1.7 Operators

Operators are actions on quantum states that transform them.

For a quantum state represented by ket |ψ>, an operator Â can act on it and transform it into another quantum state represented by ket |ϕ⟩.

$$\hat{A}|\Psi\rangle = |\phi\rangle$$

Example: Suppose

$$|\Psi\rangle = |0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$$

Suppose an operator $\hat{X} = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$

$$\hat{X}|\Psi\rangle = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix} = |\phi\rangle$$

1.8 Identity Operator

The identity operator is an operator which, operating on a function, leaves the function unchanged $\boldsymbol{I}|\Psi\rangle = |\Psi\rangle$ where $I = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$$|\Psi\rangle = |0\rangle = \begin{bmatrix}1\\ 0\end{bmatrix}$$

Suppose an operator I operates on $|\Psi\rangle$

$$I|\Psi\rangle = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1\\ 0\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix} = |\Psi\rangle$$

$$I|\Psi\rangle = |\Psi\rangle = |0\rangle$$

1.9 Conjugate matrices:

If the elements in a matrix A are complex numbers, then the matrix obtained by the corresponding conjugate complex elements is called the conjugate of A and is denoted by A^* .

Suppose $A = \begin{bmatrix}0 & i\\ -i & 0\end{bmatrix}$ therefore the complex conjugate of the matrix A can be written as $A^* = \begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}$

1.10 Transpose Matrices

If columns and rows of a matrix A are interchanged then the resultant matrix is transpose of A and represented as A^T .

Suppose $A = \begin{bmatrix}0 & 1\\ -i & 0\end{bmatrix}$ therefore the Transpose of the matrix A can be written as $A^T = \begin{bmatrix}0 & -i\\ 1 & 0\end{bmatrix}$

1.11 Hermitian Matrices

The transpose of complex conjugate of a matrix is known as Hermitian operator and the resultant matrix is known as Hermitian matrix if $A = A^{*T} = \boldsymbol{A}^{\dagger}$. It is represented by $\boldsymbol{A}^{\dagger}$

Suppose $A = \begin{bmatrix}8 & 1+i\\ 1-i & 5\end{bmatrix}$ therefore the complex conjugate of matrix A is $A^* = \begin{bmatrix}8 & 1-i\\ 1+i & 5\end{bmatrix}$ and Transpose of the matrix A^* can be written as $A^{*T} = \begin{bmatrix}8 & 1+i\\ 1-i & 5\end{bmatrix}$

Therefore A is called the Hermitian matrix if $A = A^{*T} = \boldsymbol{A}^{\dagger}$

1.12 Unitary Matrix

A matrix is said to be Unitary if

$$A\boldsymbol{A}^{\dagger} = I$$

1.13 Pauli Matrices

$$\sigma_x = X = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} \qquad \sigma_z = Z = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$$

$$\sigma_y = Y = \begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}$$

1.14 Properties of Pauli's Matrices

(a) Square Pauli matrices gives identity matrix I

$$X^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$X^2 = I$$

Similarly

$$Y^2 = I$$

$$Z^2 = I$$

(b) Pauli matrices are unitary matrix

$$X\boldsymbol{X}^\dagger = I$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Similarly

$$Y\boldsymbol{Y}^\dagger = I$$

$$Z\boldsymbol{Z}^\dagger = I$$

(c) Pauli matrices are Hermitian:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\boldsymbol{X}^\dagger = X^{*T} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$X = \boldsymbol{X}^\dagger$ **Similarly**

$$Y = \boldsymbol{Y}^\dagger$$

$$Z = \boldsymbol{Z}^\dagger$$

1.15 Quantum Gates:

A quantum gate is a very simple computing device that performs quantum operation on qubits. Quantum gates are one of the essential parts of a quantum computer and are the building blocks of all quantum algorithms.

Quantum gates are mathematically represented as transformation matrices which operate on inputs to give outputs.

There are different types of quantum gates. Single-qubit gates and multiple qubit gates. These gates can flip a qubit from 0 to 1 as well as allowing superposition states to be created.

1.16 Single-Qubit Gates:

As we have studies earlier Single Qubits are $|0\rangle and\ |1\rangle$.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}\ and\ |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Single Qubit Gates are X- gate, Y-gate, Z- gate, H –gate, S-gate, T-gate.

(i) X – Gate or Quantum Not Gate

X-gate is the single qubit input gate and is also called as Pauli X – Gate or quantum NOT Gate.

The Matrix form of X is given by $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Operation of X- Gate on $|0\rangle, |1\rangle, and\ |\Psi\rangle$ can be evaluated as following.

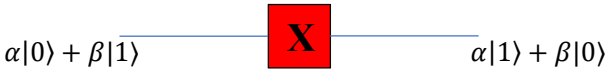
$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

$$\Rightarrow \mathbf{X|0\rangle = |1\rangle}$$

Similarly X|1⟩=|0⟩

Since X inverts each input it is also called as **bit-flip gate**.

If $|\Psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ therefore $X|\Psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix} = \alpha|1\rangle + \beta|0\rangle$



X-Gate Truth Table

X-Gate Truth Table	
Input	**Output**
$\mathbf{|0\rangle}$	$\mathbf{|1\rangle}$
$\mathbf{|1\rangle}$	$\mathbf{|0\rangle}$
$\boldsymbol{\alpha|0\rangle + \beta|1\rangle}$	$\boldsymbol{\alpha|1\rangle + \beta|0\rangle}$

(ii) Y – Gate or Pauli Y – gate.

The matrix form of Y gate is $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

Operations of Y gate on $|0\rangle, and\ |1\rangle$ provides following output

$$\boldsymbol{Y|0\rangle = i|1\rangle}$$

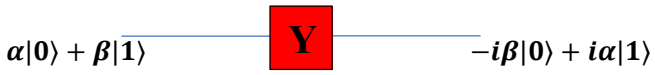
$$\Rightarrow \boldsymbol{Y|0\rangle} = \begin{bmatrix} \mathbf{0} & \boldsymbol{-i} \\ \boldsymbol{i} & \mathbf{0} \end{bmatrix}\begin{bmatrix} \mathbf{1} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{i} \end{bmatrix} = \boldsymbol{i}\begin{bmatrix} \mathbf{0} \\ \mathbf{1} \end{bmatrix} = \boldsymbol{i}|\mathbf{1}\rangle$$

Similarly $\boldsymbol{Y|1\rangle} = \begin{bmatrix} \mathbf{0} & \boldsymbol{-i} \\ \boldsymbol{i} & \mathbf{0} \end{bmatrix}\begin{bmatrix} \mathbf{0} \\ \mathbf{1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{-i} \\ \mathbf{0} \end{bmatrix} = -\boldsymbol{i}\begin{bmatrix} \mathbf{1} \\ \mathbf{0} \end{bmatrix}$

$$\boldsymbol{Y|1\rangle = -\,i|0\rangle}$$

If a superposed Qubit goes through Y-Gate

If $|\Psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ therefore $\boldsymbol{Y|\Psi\rangle} = \begin{bmatrix} \mathbf{0} & \boldsymbol{-i} \\ \boldsymbol{i} & \mathbf{0} \end{bmatrix}\begin{bmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{bmatrix} = \begin{bmatrix} \boldsymbol{-i\beta} \\ \boldsymbol{i\alpha} \end{bmatrix} = \boldsymbol{-i\beta|0\rangle + i\alpha|1\rangle}$



Y-Gate Truth Table

Y-Gate Truth Table	
Input	**Output**
$\|0\rangle$	$i\|1\rangle$
$\|1\rangle$	$-i\|0\rangle$
$\alpha\|0\rangle + \beta\|1\rangle$	$i\alpha\|1\rangle - i\beta\|0\rangle$

(iii) Z – Gate : Z-gate is the single qubit input gate. This is also called as Pauli Z – gate.

The matrix form of Z gate is $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Action of Z gate on inputs: When Z operates on $|0\rangle$ and $|1\rangle$ the phase will change. Hence this is also called as phase-flip gate

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

$$Z|1\rangle = -|1\rangle$$

$$Z|\Psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix} = \alpha|0\rangle - \beta|1\rangle$$

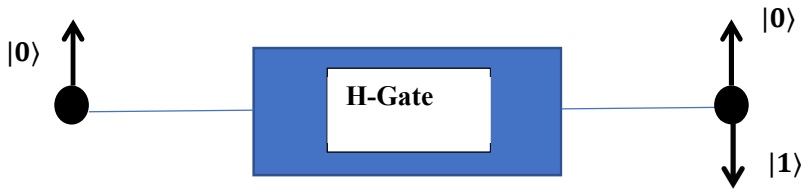
$$Z|\Psi\rangle = \alpha|0\rangle - \beta|1\rangle$$



Z-Gate Truth Table

Z-Gate Truth Table	
Input	**Output**
$\|0\rangle$	$\|0\rangle$
$\|1\rangle$	$-\|1\rangle$
$\alpha\|0\rangle + \beta\|1\rangle$	$\alpha\|0\rangle - \beta\|1\rangle$

(iv) Hadamard Gate (H-gate) – It is a single qubit gate and is also gate to superposition. H gate acts on single qubit input and produce superposition state output. Matrix form of H gate and its symbol is $\boldsymbol{H} = \frac{1}{\sqrt{2}}\begin{bmatrix}\mathbf{1} & \mathbf{1}\\ \mathbf{1} & \mathbf{-1}\end{bmatrix}$

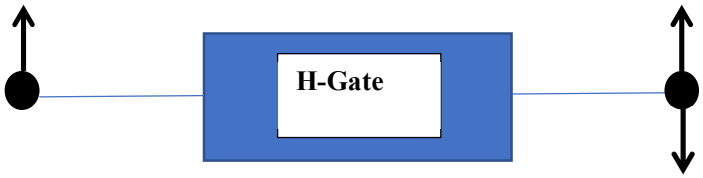


$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}\mathbf{1} & \mathbf{1}\\ \mathbf{1} & \mathbf{-1}\end{bmatrix}\begin{bmatrix}\mathbf{1}\\ \mathbf{0}\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}\mathbf{1}\\ \mathbf{1}\end{bmatrix} = \frac{1}{\sqrt{2}}(|\mathbf{0}\rangle + |\mathbf{1}\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}\mathbf{1} & \mathbf{1}\\ \mathbf{1} & \mathbf{-1}\end{bmatrix}\begin{bmatrix}\mathbf{0}\\ \mathbf{1}\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}\mathbf{1}\\ \mathbf{-1}\end{bmatrix} = \frac{1}{\sqrt{2}}(|\mathbf{0}\rangle - |\mathbf{1}\rangle)$$

$$H|\Psi\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}\mathbf{1} & \mathbf{1}\\ \mathbf{1} & \mathbf{-1}\end{bmatrix}\begin{bmatrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}\boldsymbol{\alpha}+\boldsymbol{\beta}\\ \boldsymbol{\alpha}-\boldsymbol{\beta}\end{bmatrix} = \frac{(\boldsymbol{\alpha}+\boldsymbol{\beta})}{\sqrt{2}}|\mathbf{0}\rangle + \frac{(\boldsymbol{\alpha}-\boldsymbol{\beta})}{\sqrt{2}}|\mathbf{1}\rangle$$

$$H|\Psi\rangle = \alpha\frac{(|\mathbf{0}\rangle+|\mathbf{1}\rangle)}{\sqrt{2}} + \boldsymbol{\beta}\frac{(|\mathbf{0}\rangle-|\mathbf{1}\rangle)}{\sqrt{2}}$$



$|\Psi\rangle = \alpha|\mathbf{0}\rangle + \boldsymbol{\beta}|\mathbf{1}\rangle$ $\qquad$ $\alpha\frac{(|\mathbf{0}\rangle+|\mathbf{1}\rangle)}{\sqrt{2}} + \boldsymbol{\beta}\frac{(|\mathbf{0}\rangle-|\mathbf{1}\rangle)}{\sqrt{2}}$

H-Gate Truth Table

H-Gate Truth Table	
Input	**Output**
$\|\mathbf{0}\rangle$	$\frac{(\|\mathbf{0}\rangle + \|\mathbf{1}\rangle)}{\sqrt{2}}$
$\|\mathbf{1}\rangle$	$\frac{(\|\mathbf{0}\rangle - \|\mathbf{1}\rangle)}{\sqrt{2}}$
$\boldsymbol{\alpha}\|\mathbf{0}\rangle + \boldsymbol{\beta}\|\mathbf{1}\rangle$	$\alpha\frac{(\|\mathbf{0}\rangle + \|\mathbf{1}\rangle)}{\sqrt{2}} + \boldsymbol{\beta}\frac{(\|\mathbf{0}\rangle - \|\mathbf{1}\rangle)}{\sqrt{2}}$

Note: Difference between X, Y, Z and H gates is that in X, Y and Z gates, output is in single state whereas in H gate output is superposed state.